Proving the Wrong Thing, Faster

Kacper Bak, PhD

Code ✓

Tests ✓

Proof ✓

Intent ?

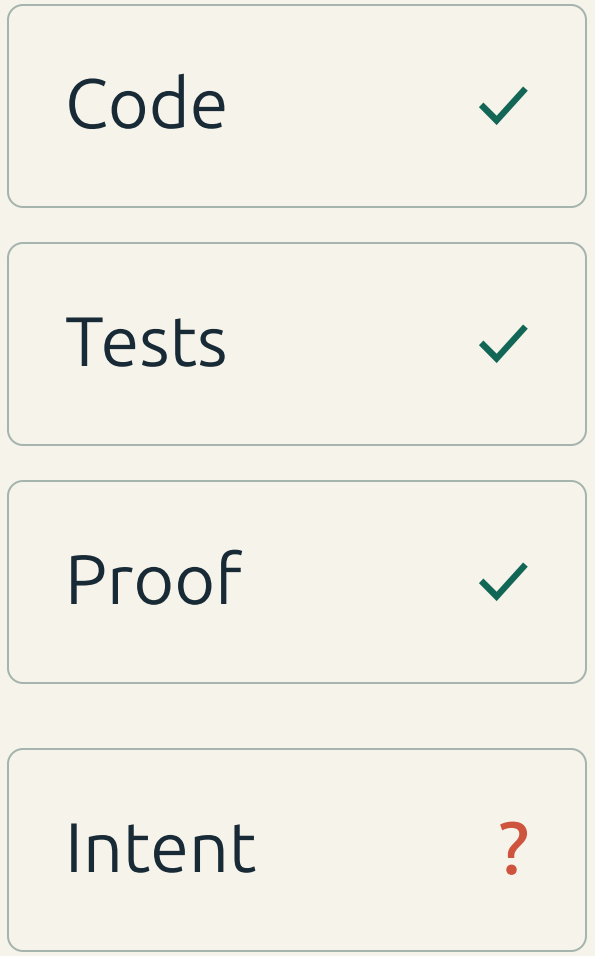
Keep verifying the right things.

Shared misunderstanding.

A valid proof of the wrong property.

Change reopens the question.

How to keep artifacts aligned with the intent?



IntentBond helps agents preserve intent.

Proving a system is “safe” depends on what we mean by “safe.”

VERIFICATION

Are we building the **system right?**

Behavior against the specification

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VERIFICATION

Are we building the **system** **right?**

Behavior against the specification

VALIDATION

Are we building the **right** **system?**

The system against intended use

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VERIFICATION

Are we building the system right?

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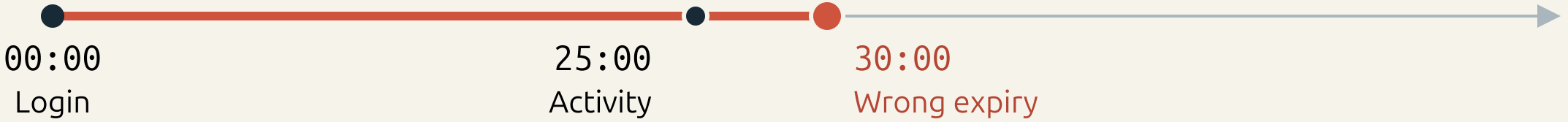
Verification can succeed while validation fails

Verification cannot decide what users need

Correct proof. Wrong promise.

“Expire session after 30 minutes of **inactivity.**”

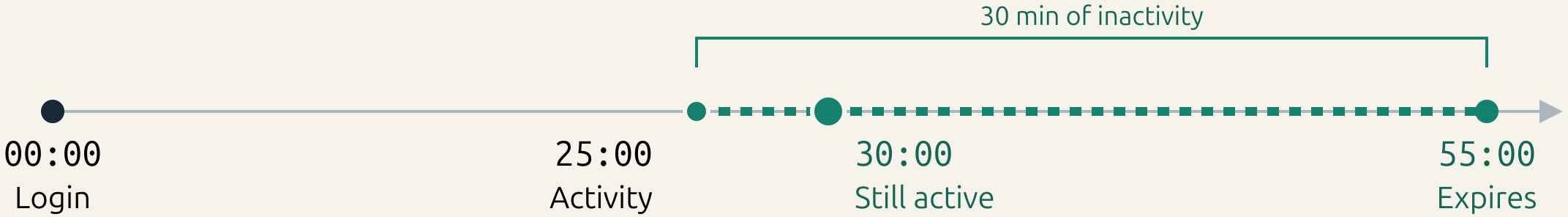
Agent:




```
expired ↔ now − login_time ≥ 30 min
```

The intended behavior.

“Expire session after 30 minutes of **inactivity.**”




```
expired ↔ now − last_activity ≥ 30 min
```

Is your agent still building the right system?

Next request: “Add audit logging to session expiry.”

Invent	Add a session lifetime cap nobody requested.
Miss	Forget that activity resets the timer.
Reinterpret	Count background heartbeats as user activity.
Delete	Remove session expiry.

Tests and proofs can pass against a weakened specification.

> **How to keep intent, spec, implementation, tests, and evidence aligned?**

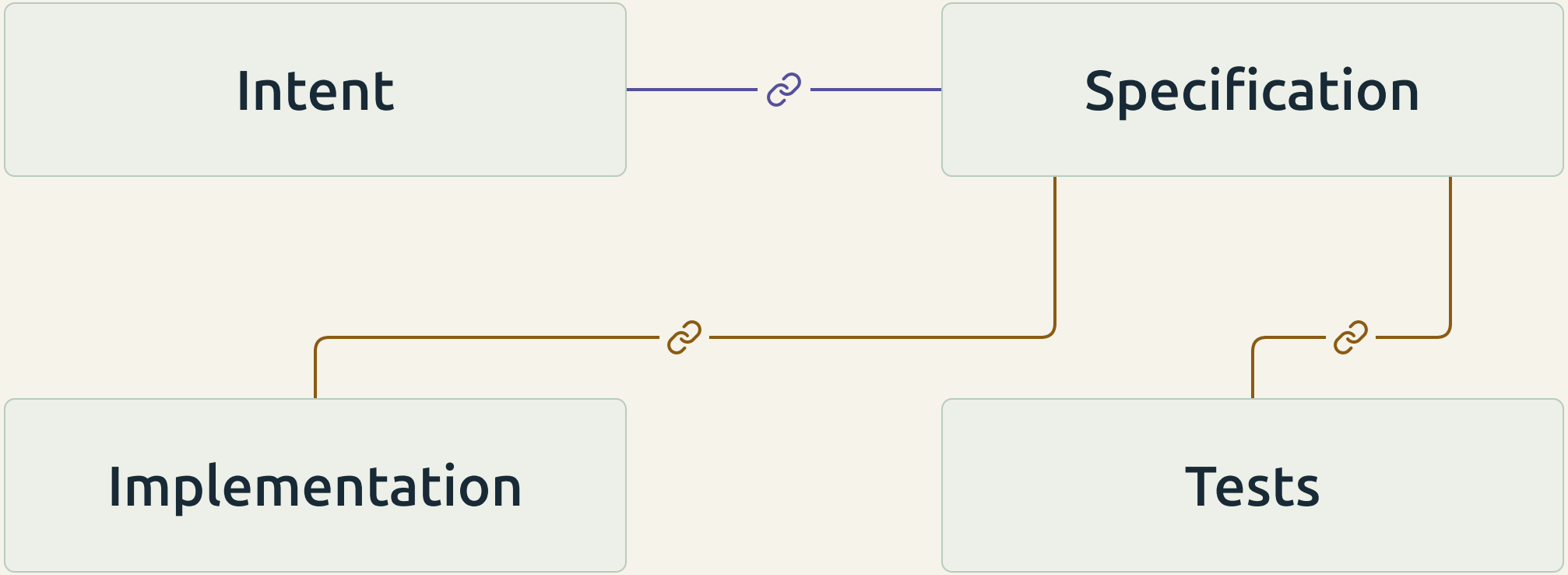
IntentBond keeps development artifacts in sync.

Agent skills + tools: links, checks, guidance for dev workflow

Connect	Establish links between intent, spec, implementation, and tests.
Guide	Direct repairs with deterministic feedback.
Validate	Review behavior against intent; repair or clarify.
Refine	Gradually formalize requirements into properties and models.

Built on OpenFastTrace.
Related: Covmap · Tracey · Kibi · Code Contracts · OpenSpec

Connect: links between artifacts.



Document split follows Anthropic's AI-native SDLC playbook.

Connect: links between artifacts.

intent.md — Desired outcome

`intent~abandoned-sessions`

User should not leave an unattended account accessible indefinitely.

Needs: req

spec.md — Concrete requirement

`req~session-expiration`

Expire after 30 minutes of inactivity.
expired ↔ inactivity ≥ 30 min

Covers:
- `intent~abandoned-sessions`

Needs: impl, utest

Session.lean

```
-- [impl->req~session-expiration]
def expired (idle : Nat) : Bool :=
  decide (idle ≥ 30 * 60)
```

SessionTests.lean

```
-- [utest->req~session-expiration]
#guard expired 1799 == false
#guard expired 1800 == true
```

Document split follows Anthropic's AI-native SDLC playbook.

Guide: give agents deterministic feedback.

1 Agent edits

spec.md

`req~timeout~1` ← revision

Expire privileged sessions
after **30 minutes** idle.

Session.lean

```
-- [impl->req~timeout~1]
```

Guide: give agents deterministic feedback.

1 Agent edits

spec.md

`req~timeout~2` ← revision

Expire privileged sessions
after **5 minutes** idle.

Session.lean

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Guide: give agents deterministic feedback.

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Expire privileged sessions
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2 First IntentBond check

Rejected

Link target not found:

`req~timeout~1`

Tests passed

Guide: give agents deterministic feedback.

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3 Agent repairs

Review the revised promise.

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Guide: give agents deterministic feedback.

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- -- [impl->req~timeout~1]
+ -- [impl->req~timeout~2]
```

4 Check again after repair

Links + tests **passed**

Impact analysis across artifacts.

Coverage: detect traceability gaps.

Validate: does the change preserve intent?

IntentBond-guided review

Code against (linked) intent; ask when unclear

Current

All sessions

30 min

Validate: does the change preserve intent?

IntentBond-guided review

Code against (linked) intent; ask when unclear

Current

All sessions

30 min

Requested

Privileged sessions

5 min

Validate: does the change preserve intent?

IntentBond-guided review

Code against (linked) intent; ask when unclear

Current	Requested	Changed
All sessions	Privileged sessions	All sessions
30 min	5 min	5 min

Validate: does the change preserve intent?

IntentBond-guided review

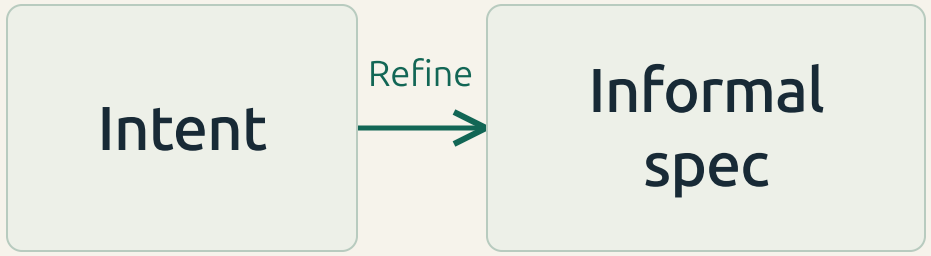
Code against (linked) intent; ask when unclear

Current	Requested	Changed
All sessions	Privileged sessions	All sessions
30 min	5 min	5 min

Agent action

Restore 30 min for ordinary sessions, then rerun checks.

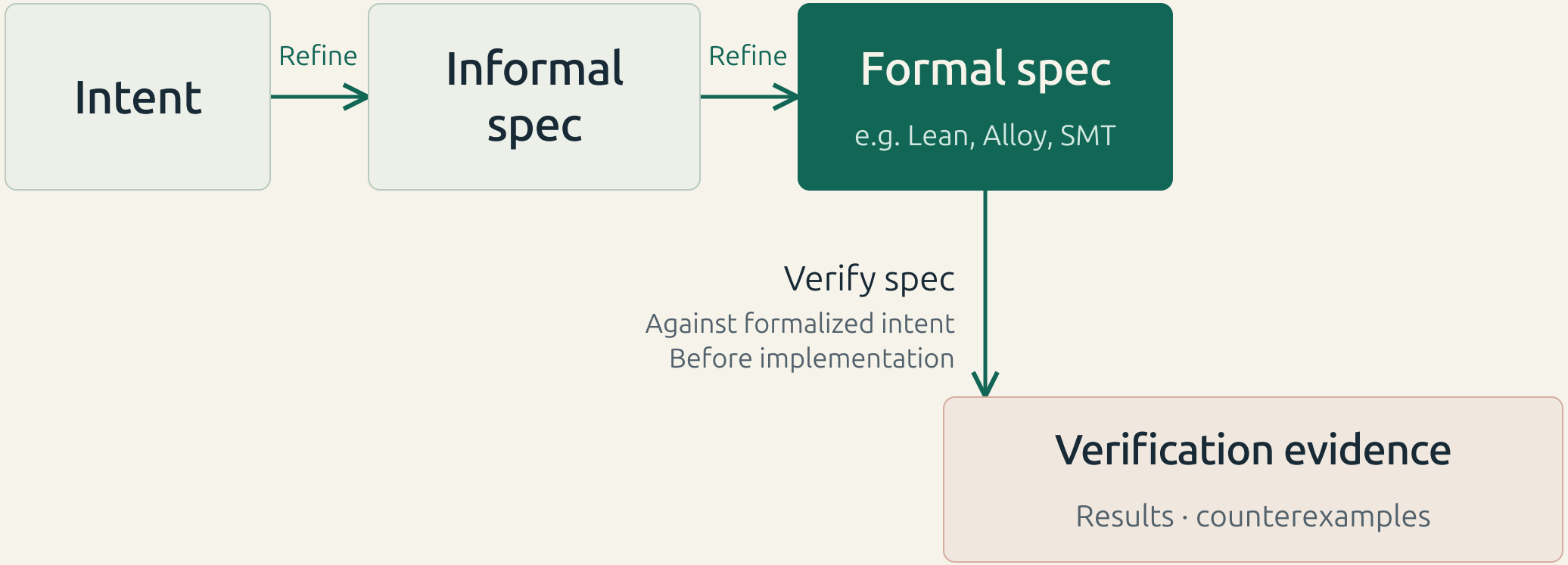
Refine: make requirements formally checkable.



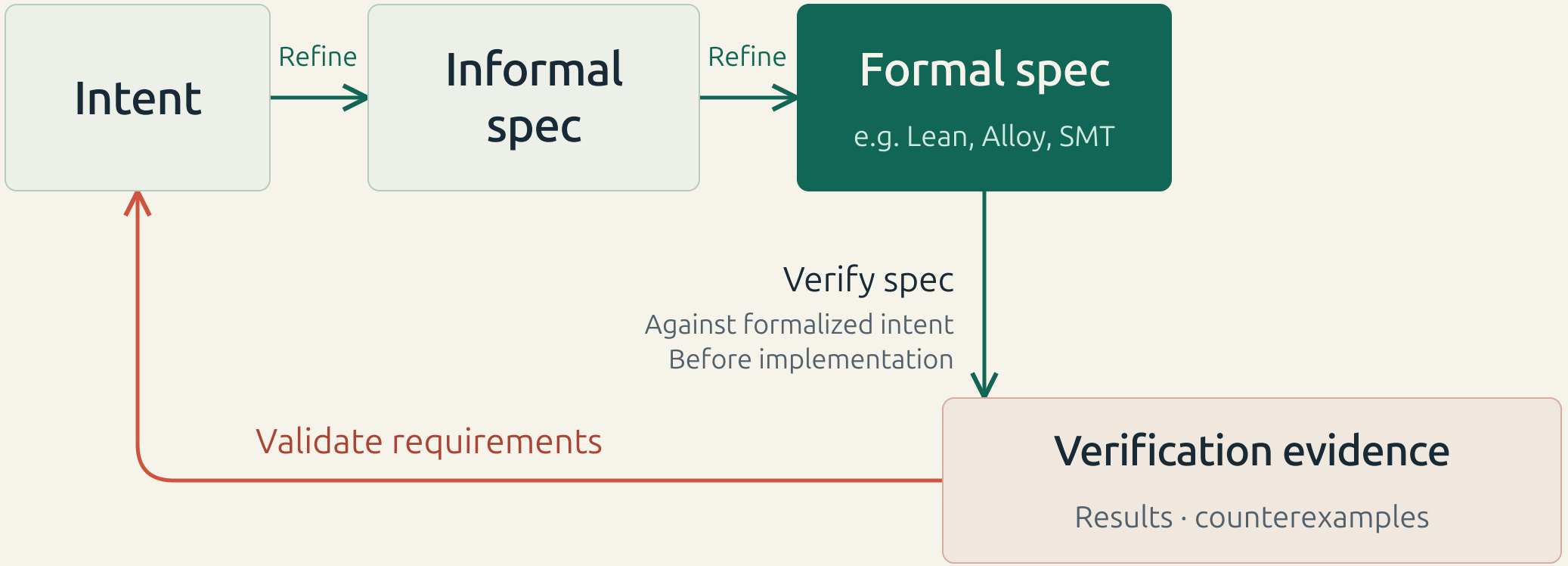
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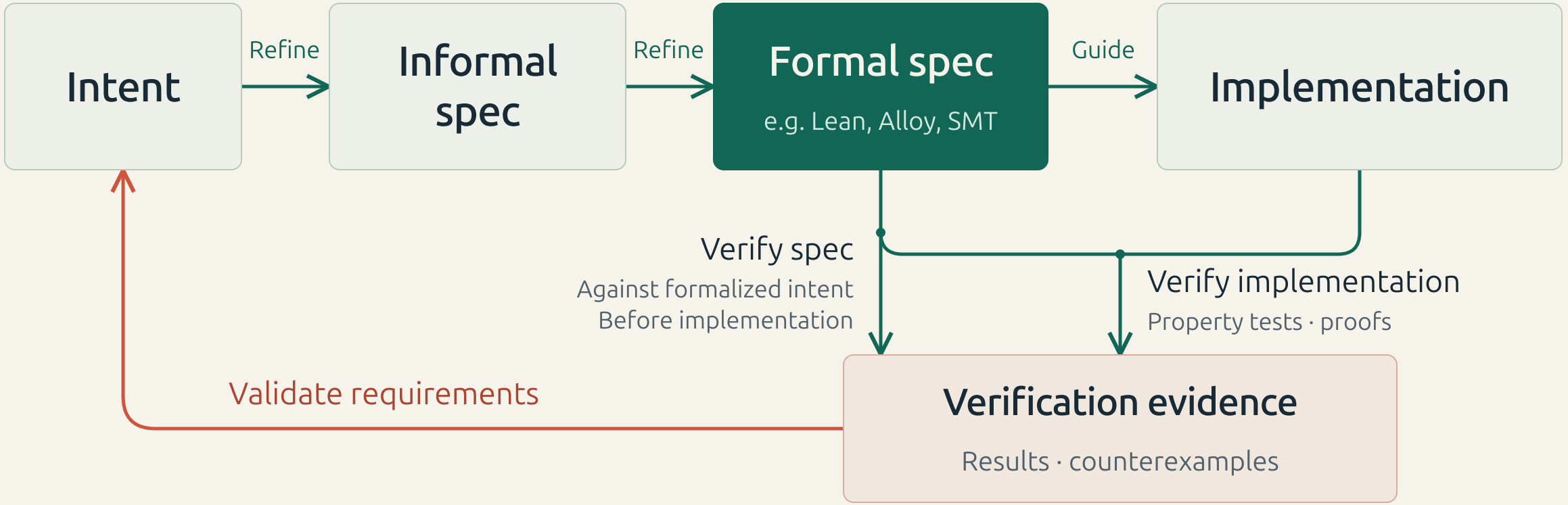
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Refine: make requirements formally checkable.



Specs can help agents write better tests.

People decide what must hold, agents help formalize it.

What’s next for IntentBond?

Empirical evaluation	Measure intent preservation, effort, and cost.
Requirements discovery	Use concrete behaviors to clarify intent.
Semantic change analysis	Detect weakened properties and changed assumptions.
Production feedback	Link intent and decisions to their observed effects.

Keep verifying the right things.

Keeping specs and code faithful to intent is still hard.

Agents help maintain alignment and decision history.

We remain responsible for the properties we choose.



github.com/kbak/intentbond

Making continuous alignment practical.

Extra slides

We don’t know every requirement upfront.

Should active sessions expire too?

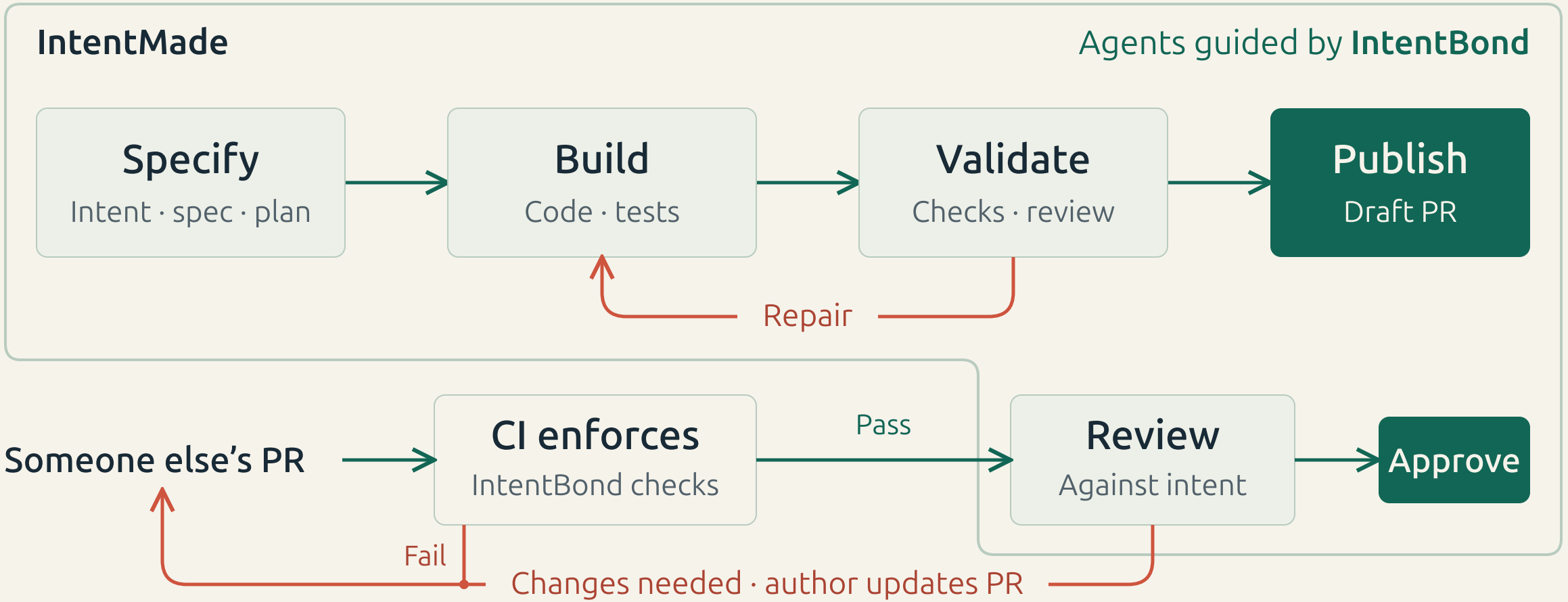
A maximum lifetime would require another condition.

Does a background heartbeat count as activity?

Counting it could keep an unattended session alive indefinitely.

AI-assisted elicitation, e.g. Matt Pocock’s grill-me

Keep alignment in the factory loop.



Publication gate: checks pass · review satisfied · evidence matches code.

Built on OFT. Packaged for agents.

Related tool	Their approach	How IntentBond differs
Covmap	Tracks mapping coverage and drift	Requires implementation and test links
Tracey	Links requirements to code/tests	Links and aligns intent with requirements
Kibi	Builds a structured knowledge base	Links existing docs, code & tests
Code Contracts	Guides agents with prose contracts	Runs checks and records evidence
OpenSpec	Agent infers correspondence	Maintains and checks versioned links

OpenFastTrace machinery + agent guidance + execution + review

Open source · inline or external mappings · standalone / CI / factories